

## **C - PROGRAM STRUCTURE**

It illustrates[Describes] how to write a program in c-language.

Every programming language is having a particular structure and we should have to follow this structure.

C-Programming structure is divided into the following parts.

- **[ documentation section ]**
- **Header files / Proto types / Preprocessor**
- **[ global variables ]**
- **[ function declarations & definitions ]**
- **void main() / main() / int main()**
- **Other statements.**

Generally documentation section consists of program headings, definitions etc and They should be represented with comments.

The statements that are enclosed in between `/*` and `*/` are called comments.

Comments never participate in program execution. They are only for user understandability or display purpose.

C-Language supports comment block only.

Eg:

```
/*  
.....;  
.....;  
*/
```

C++ supports comment block and single line comments.

Eg: `// .....`

Header files consists of function definitions, global variables, macros etc.

We can declare the header files at any place of our program. But before going to use the relevant function, its header file should be declared. It is recommended to declare the header files at the top of the program.

Every header file should be started with **#include**. Here **#** is a **preprocessor** indicator.

We can place header files in angled brackets **< >** or double quotes **" "**.

Header file never ends with **semicolon(;**).

**Note:** In C++, we should have to declare header files at the top only.

The variables that are declared before **main()** or top of the program are called **global variables** and they can be accessed from anywhere in our program. They are optional.

Function declarations and definitions contain function header and body.

- \* Every C-Program execution starts from main() function and travel towards down. Hence it is also called **top-down** approach.

- \* Without main(), C-Program never executed but compiled.

- \* main() is predefined function with user defined body. main() doesn't have any header file. One program have to maintain one main() only. **We can create alternate for main()**. Other statements are changed from program to program.

**Note:** It is recommended to write C programs in lower case only. Every statement should have to end with semicolon main()).

## **printf():**

It is a predefined function available in `stdio.h`

It is the major output function in C.

It is used to display the given text on the monitor.

In `printf`, `f` means formatted.

### **Syntax:**

```
int printf( "[text] [ conversion characters / format specifiers /  
format strings ] " [ , variables ] [ , expressions ] );
```

note:

1. In `printf` the first argument / value should be in " ".
2. In `printf` execution order is right to left but printing left to right.
3. In `printf` everything printed as it is except back slash and conversion characters.
4. `Printf` can perform both formatted and unformatted outputs.

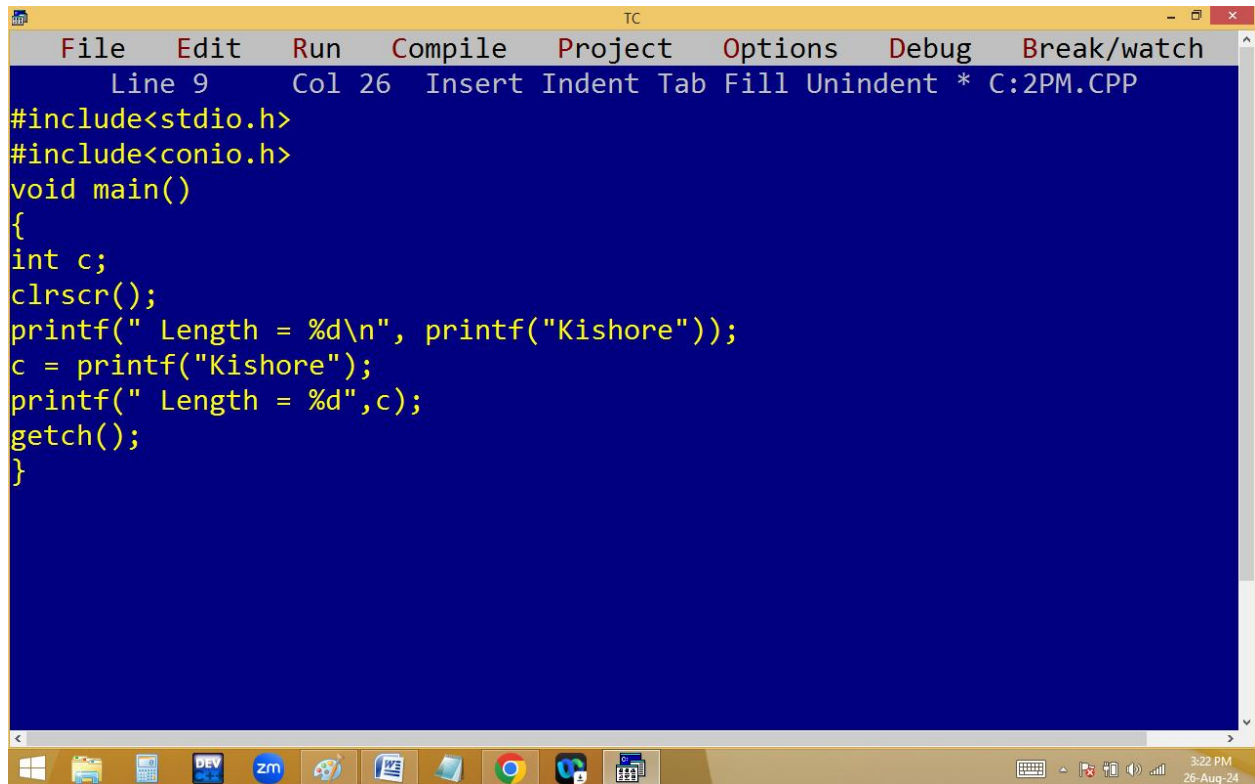
eg:

```
printf("hi"); ← unformatted
```

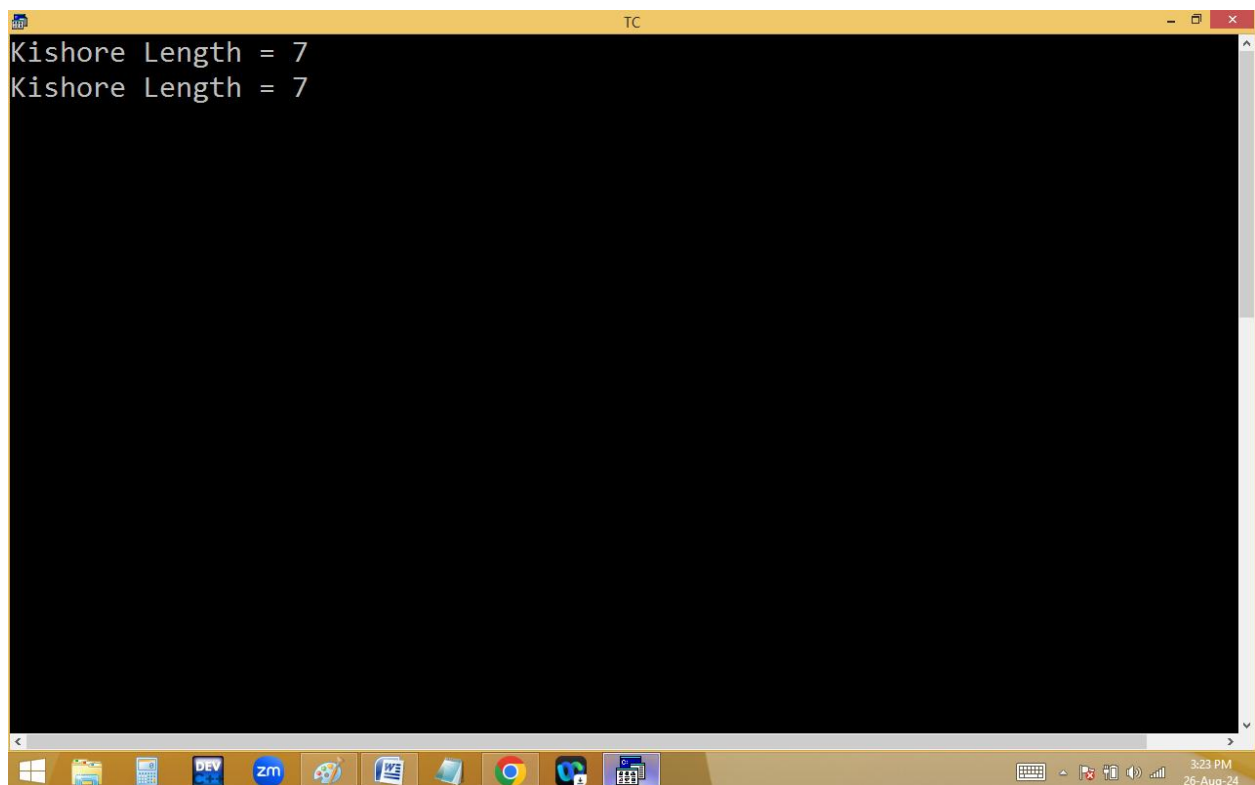
```
printf("%d",10); ← formatted
```

5. `Printf` always returns an `int` that indicates the number of printable characters.

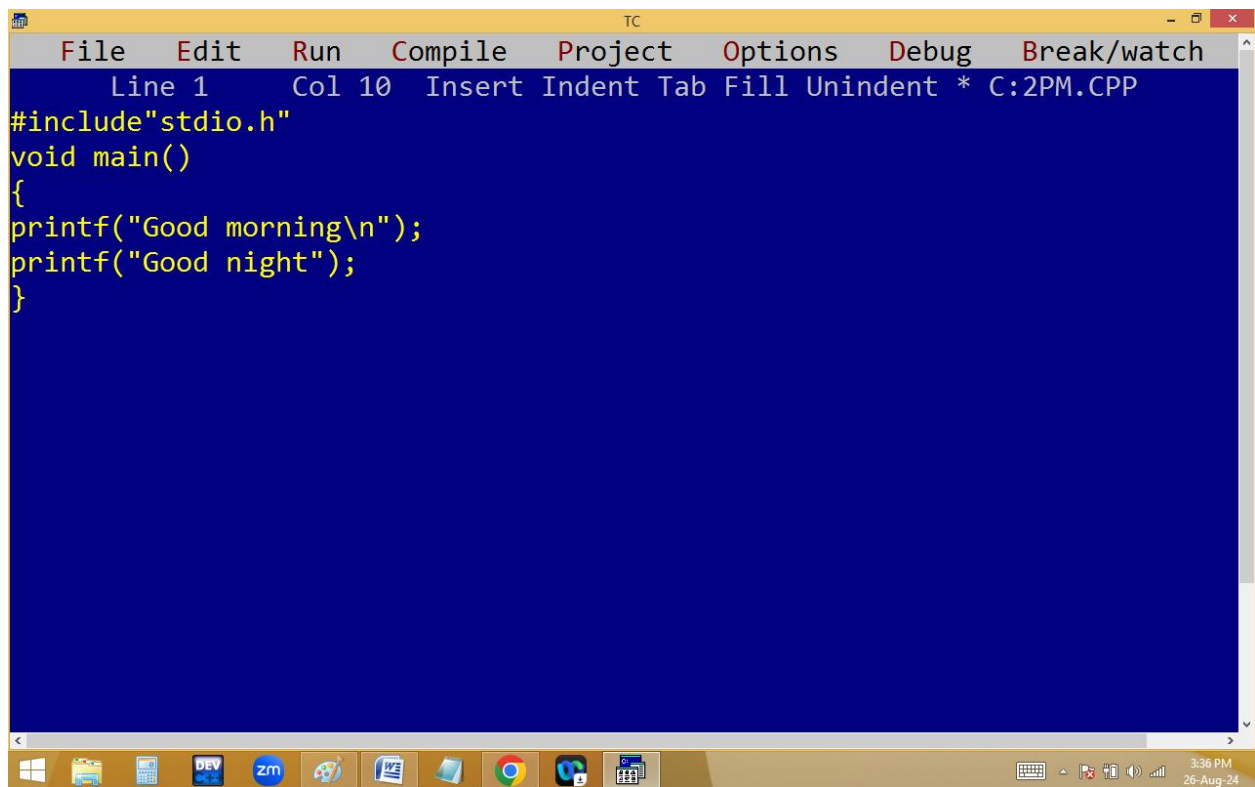
write a c program to find string length without using strlen() or a loop.



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 9 Col 26 Insert Indent Tab Fill Unindent * C:\2PM.CPP
#include<stdio.h>
#include<conio.h>
void main()
{
int c;
clrscr();
printf(" Length = %d\n", printf("Kishore"));
c = printf("Kishore");
printf(" Length = %d",c);
getch();
}
```



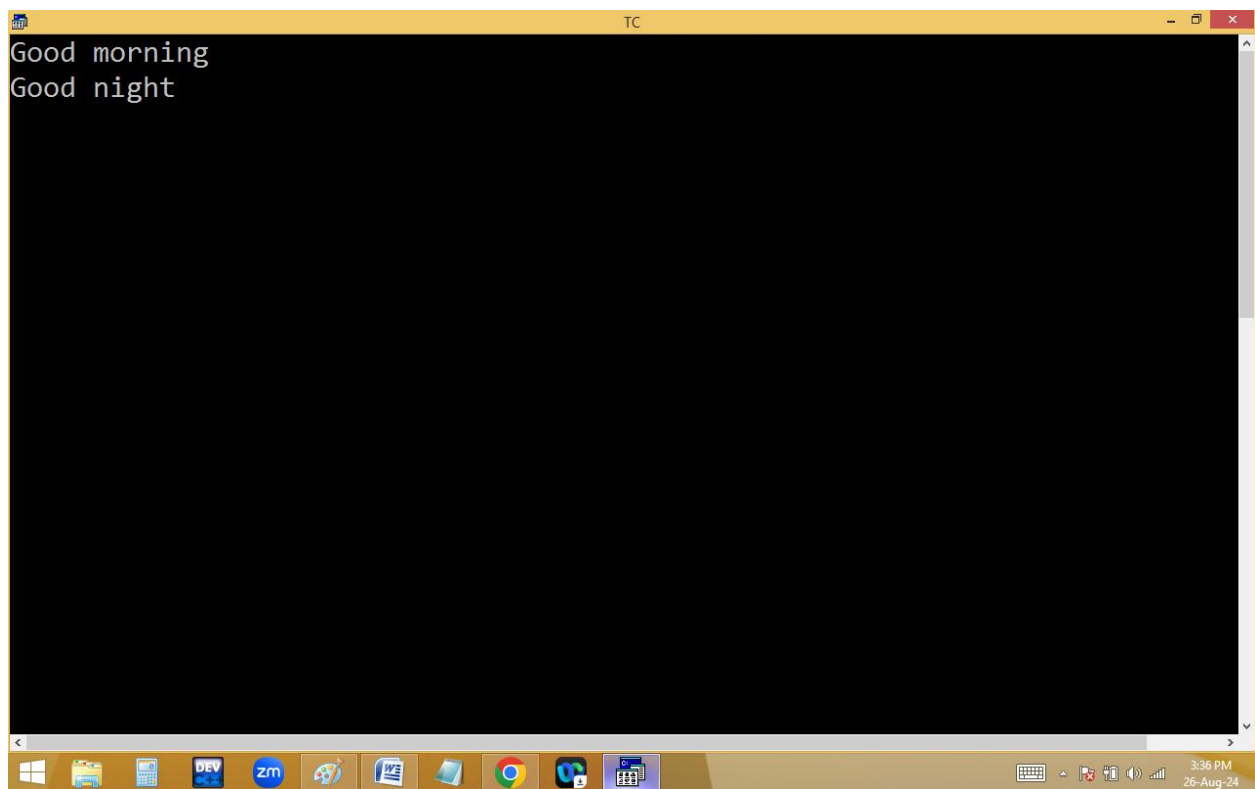
```
TC
Kishore Length = 7
Kishore Length = 7
3:23 PM
26-Aug-24
```



The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar and menu bar. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. Below the menu bar, a status bar displays 'Line 1', 'Col 10', and 'Insert Indent Tab Fill Unindent \* C:2PM.CPP'. The main editing area has a blue background and contains the following C++ code:

```
#include"stdio.h"
void main()
{
printf("Good morning\n");
printf("Good night");
}
```

The Windows taskbar at the bottom shows various application icons, including Windows Explorer, DEV, zm, and others. The system clock in the bottom right corner indicates 3:36 PM on 26-Aug-24.

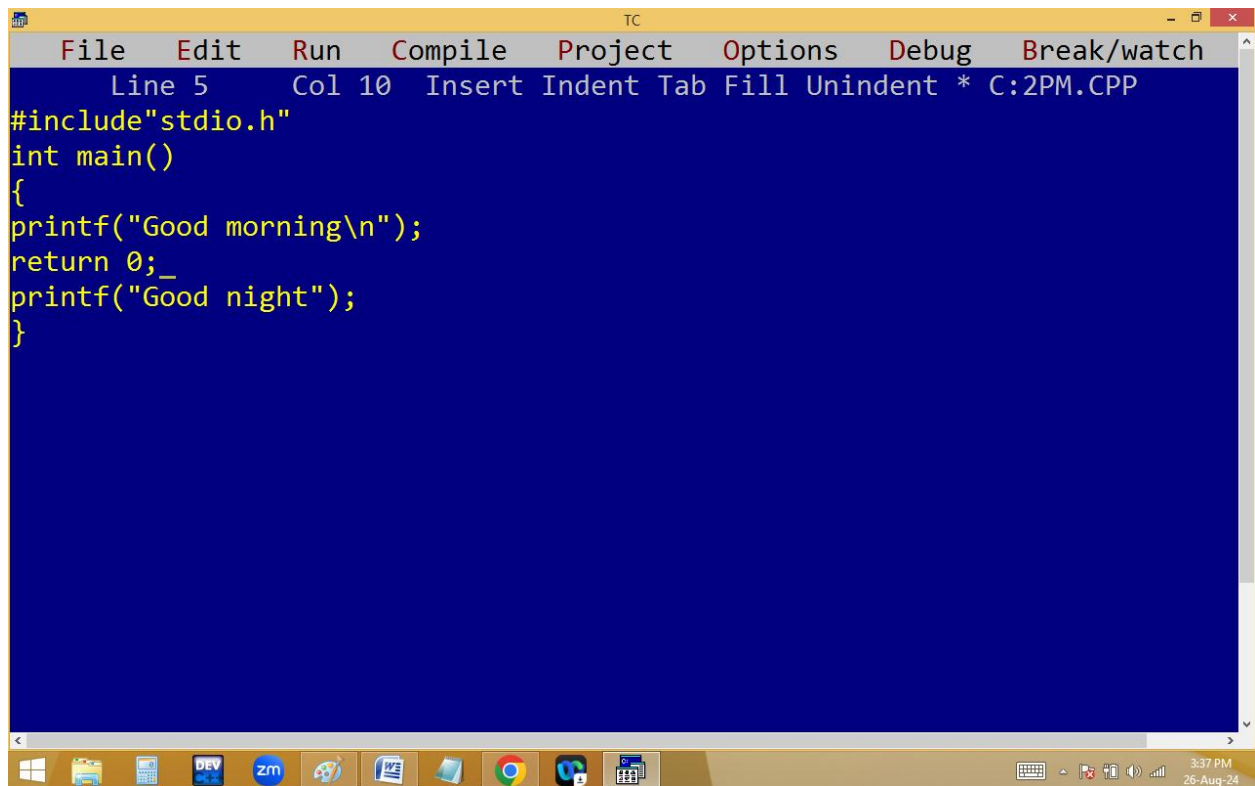


The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar and menu bar. The main editing area has a black background and displays the output of the program:

```
Good morning
Good night
```

The Windows taskbar at the bottom is identical to the first screenshot, showing the same application icons and system clock (3:36 PM on 26-Aug-24).

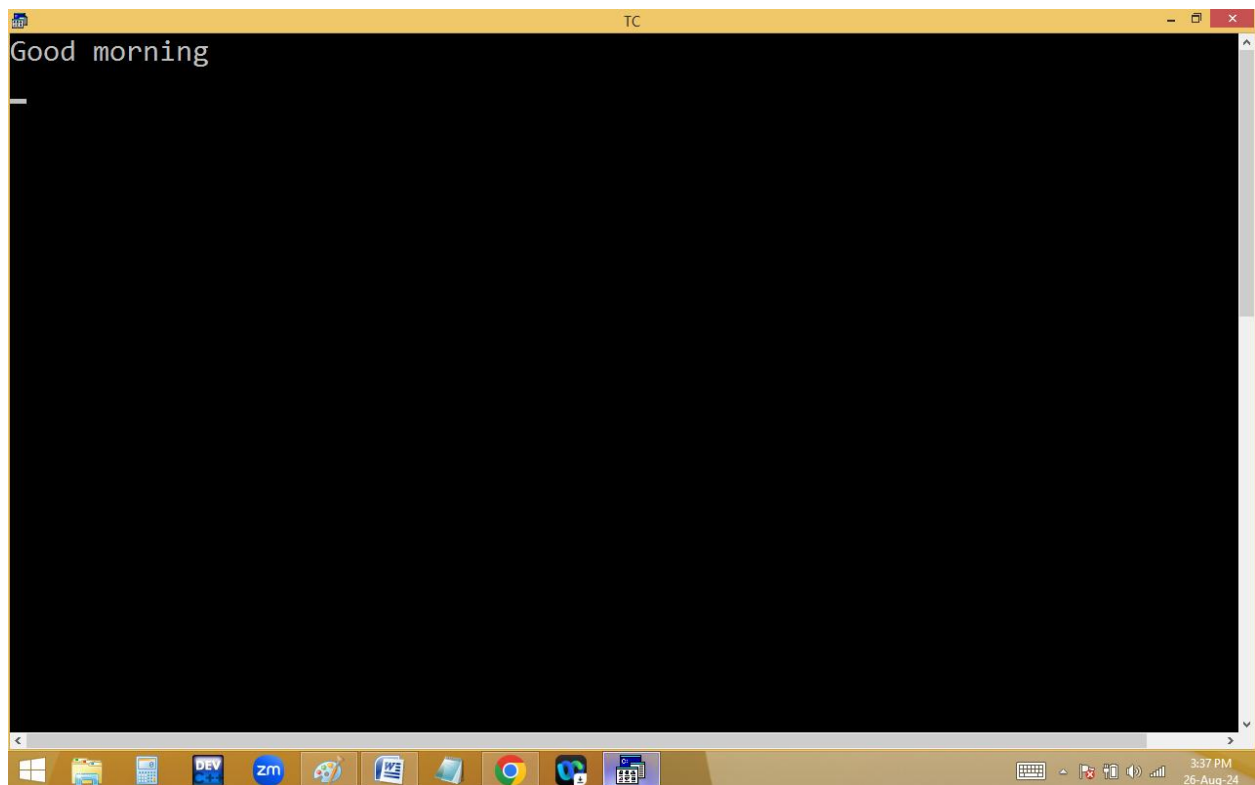




The screenshot shows the Turbo C++ (TC) IDE window. The title bar is yellow and contains the text "TC" and standard window controls. The menu bar is also yellow and includes "File", "Edit", "Run", "Compile", "Project", "Options", "Debug", and "Break/watch". Below the menu bar, a status bar displays "Line 5", "Col 10", and "Insert Indent Tab Fill Unindent \* C:2PM.CPP". The main editing area has a blue background and contains the following C code:

```
#include "stdio.h"
int main()
{
printf("Good morning\n");
return 0;_
printf("Good night");
}
```

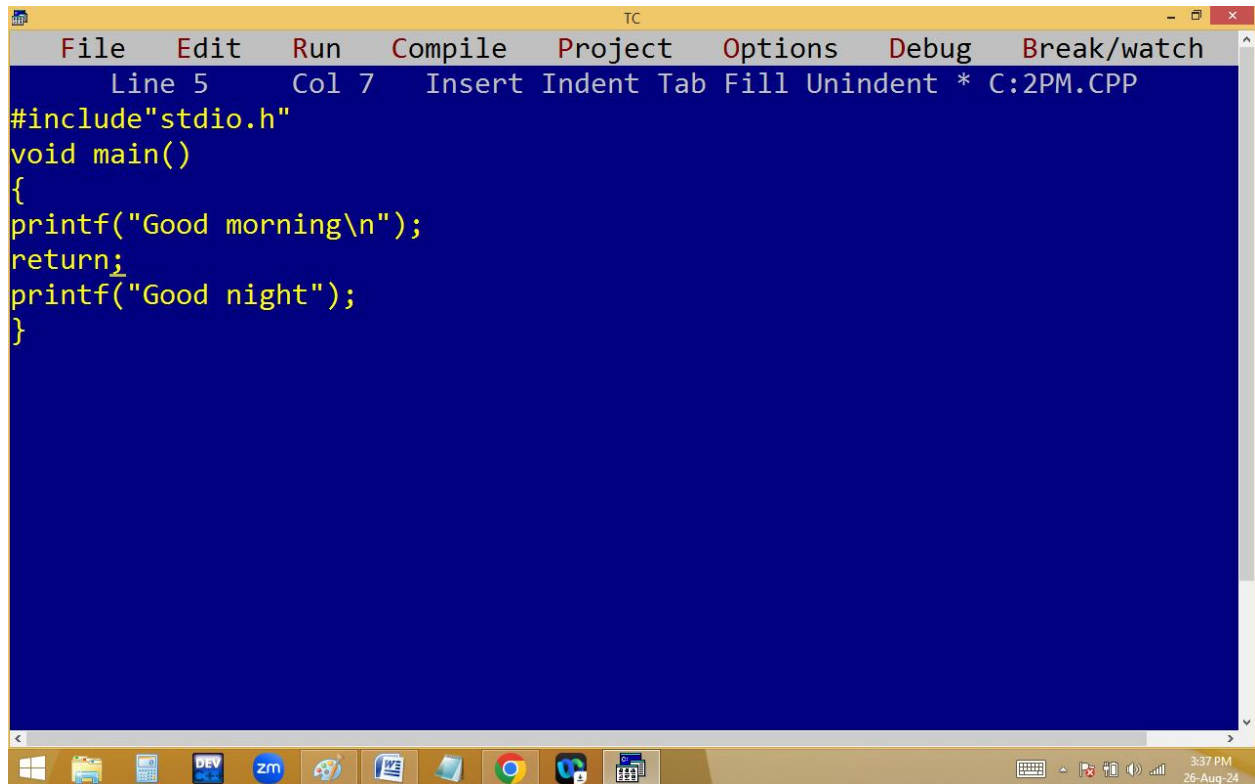
The Windows taskbar is visible at the bottom, showing icons for Windows, File Explorer, DEV, zm, a folder icon, a document icon, Google Chrome, and a calendar icon. The system tray on the right shows the time as 3:37 PM and the date as 26-Aug-24.



The screenshot shows the Turbo C++ (TC) IDE window after execution. The title bar is yellow and contains the text "TC" and standard window controls. The main editing area has a black background and displays the output of the program:

```
Good morning
```

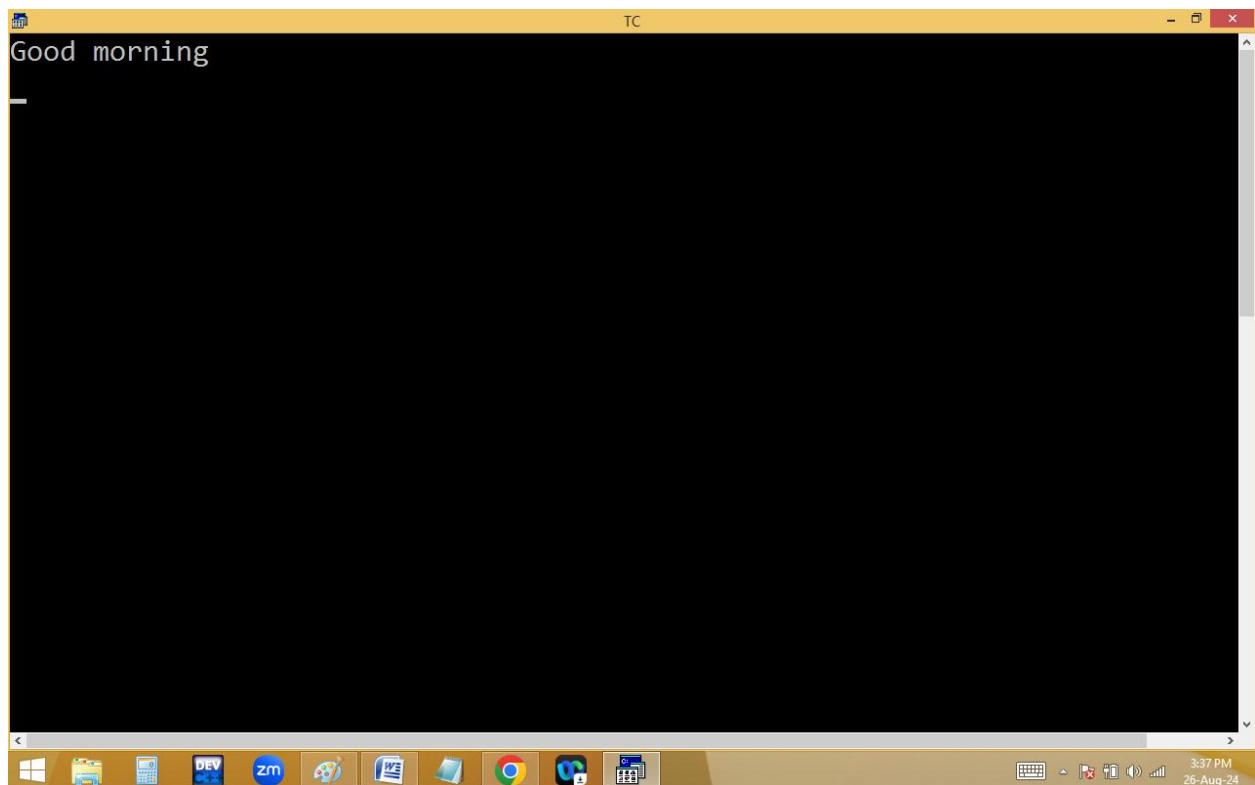
The Windows taskbar is visible at the bottom, showing icons for Windows, File Explorer, DEV, zm, a folder icon, a document icon, Google Chrome, and a calendar icon. The system tray on the right shows the time as 3:37 PM and the date as 26-Aug-24.



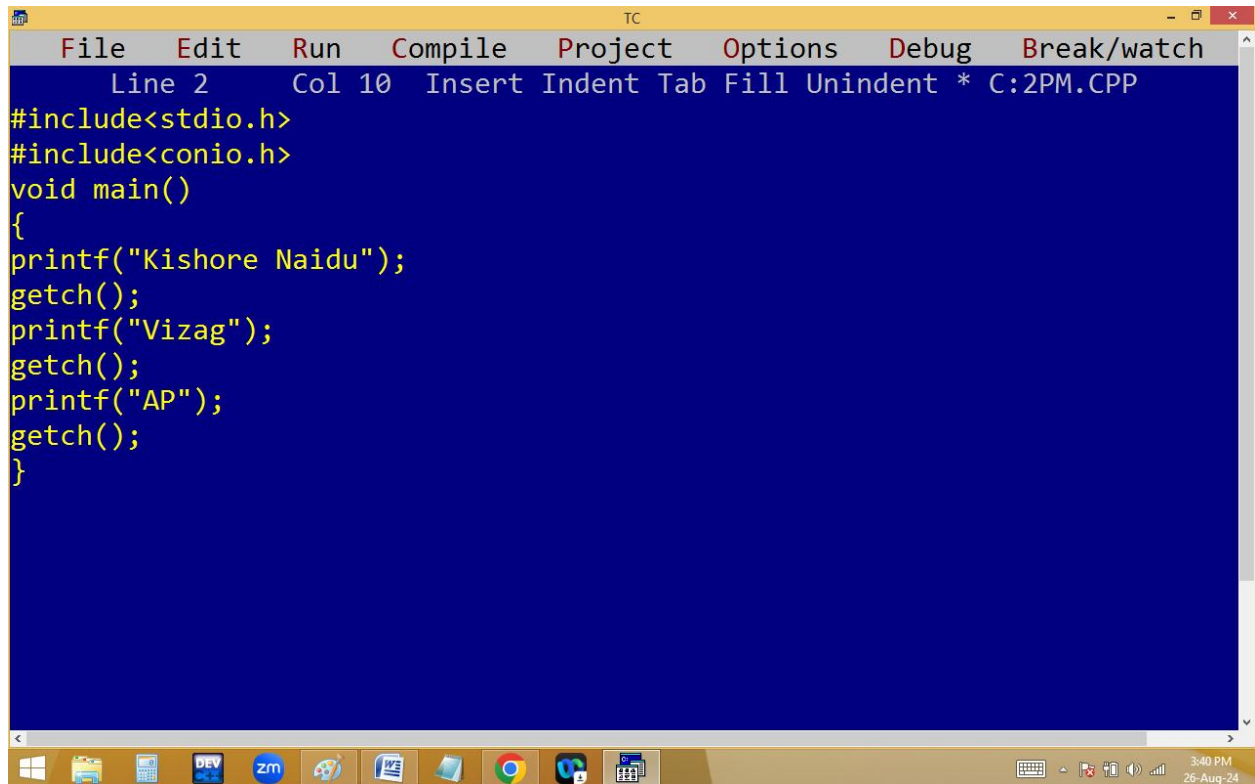
The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar and menu bar. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 5 Col 7' and the file path 'C:\2PM.CPP'. The main editing area has a blue background and contains the following C code:

```
#include "stdio.h"
void main()
{
printf("Good morning\n");
return;
printf("Good night");
}
```

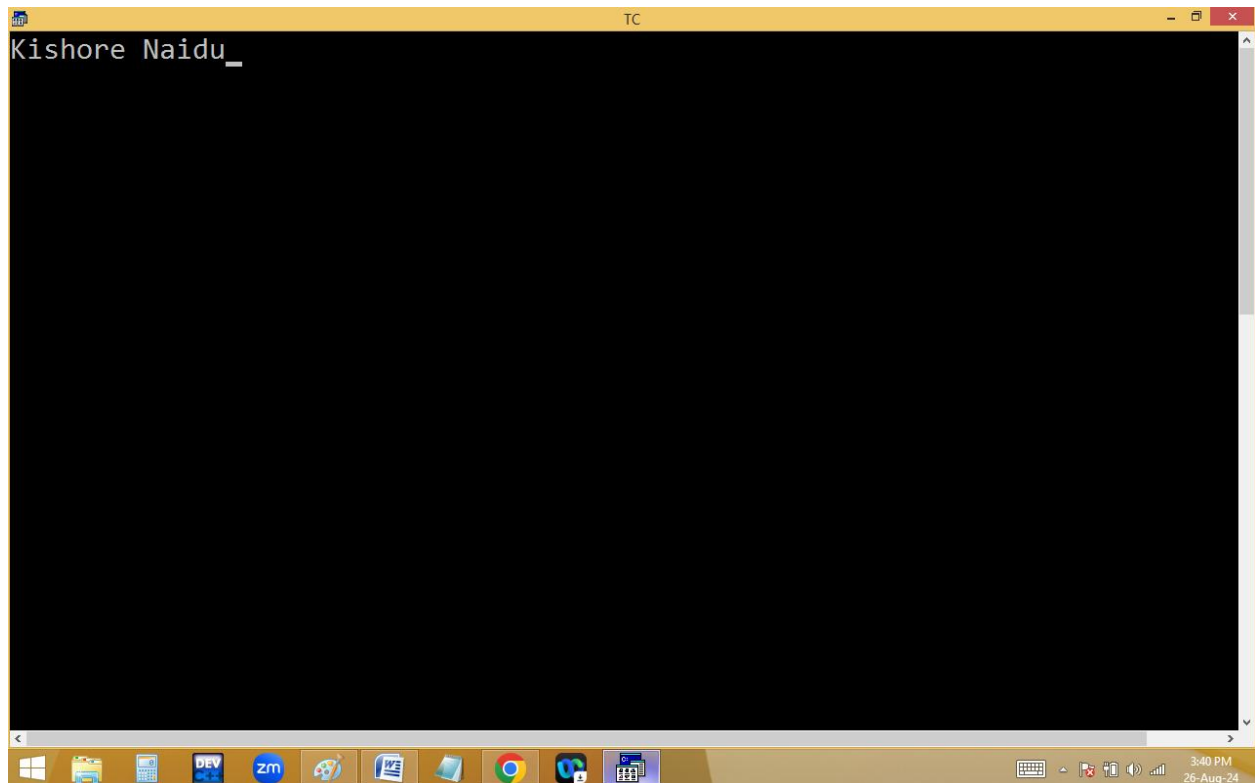
The Windows taskbar at the bottom shows various application icons and the system clock displaying 3:37 PM on 26-Aug-24.



The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar. The main editing area has a black background and displays the output of the program: 'Good morning'. The Windows taskbar at the bottom is identical to the first screenshot, showing the same application icons and system clock.

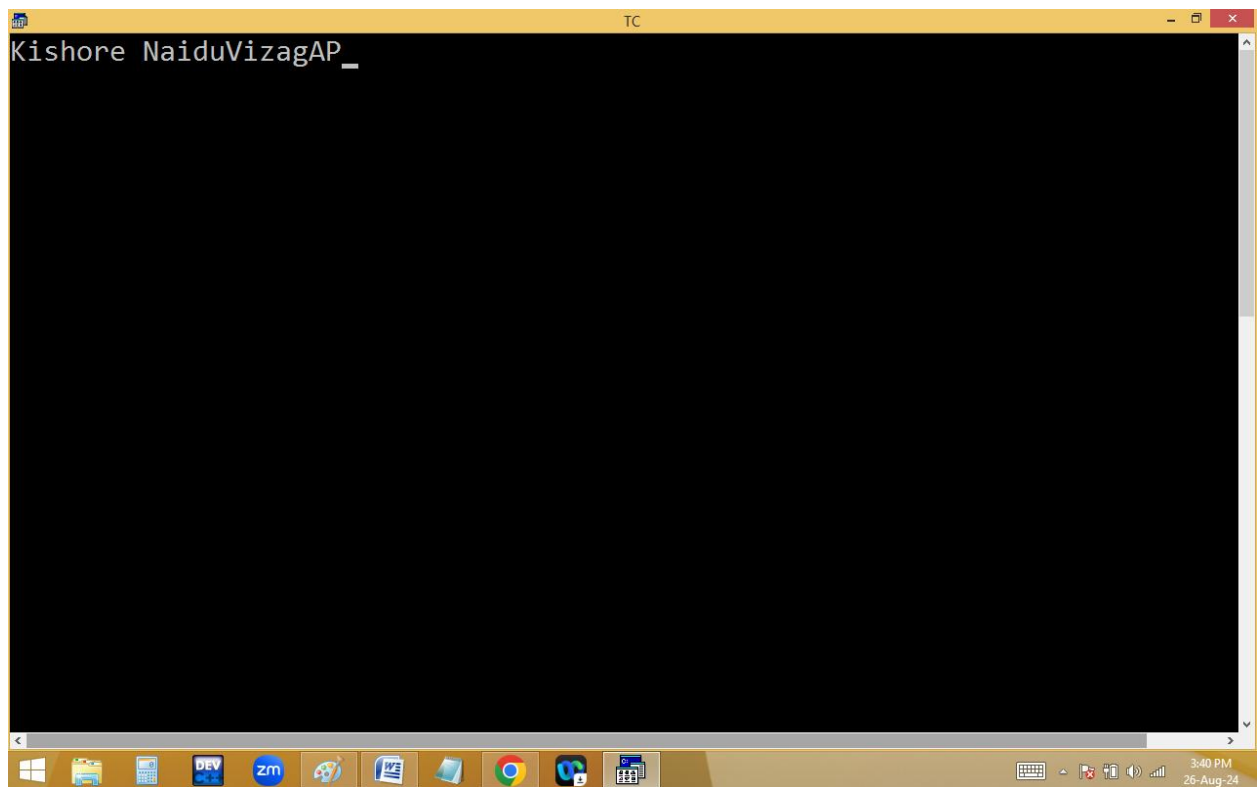
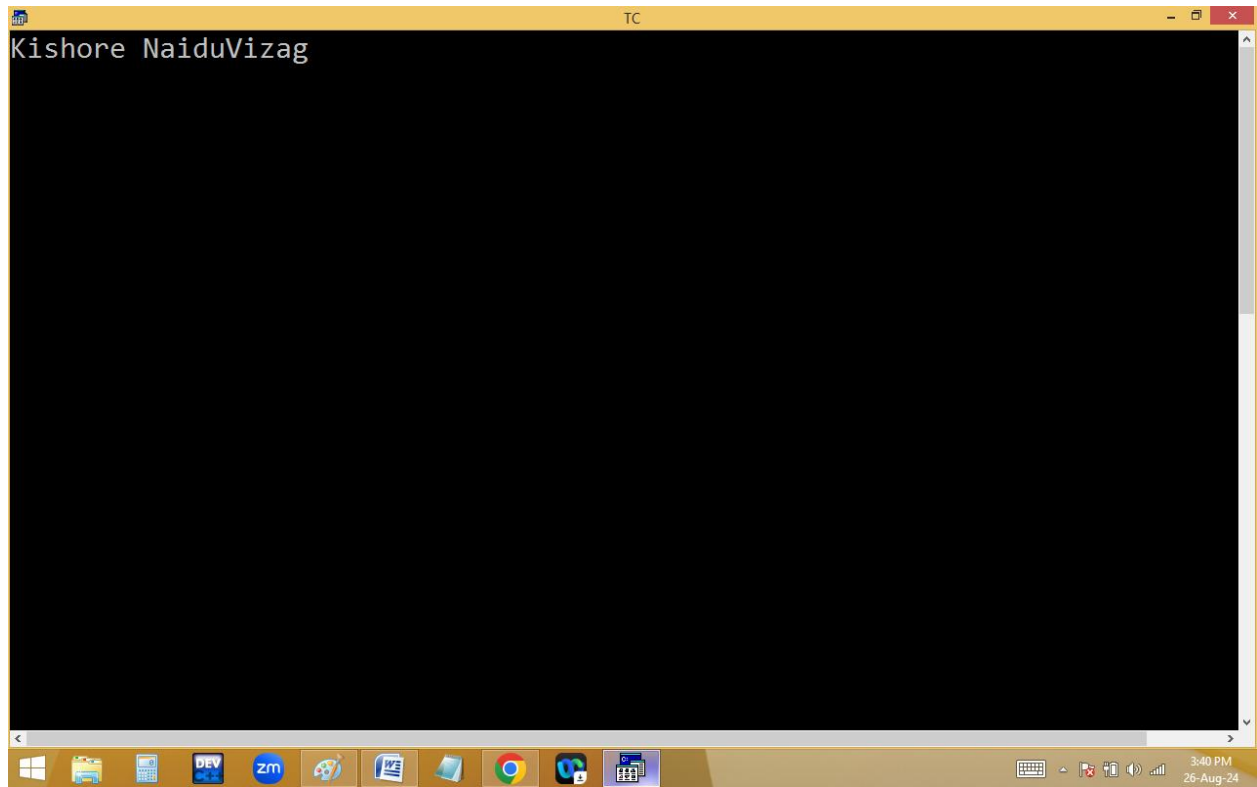


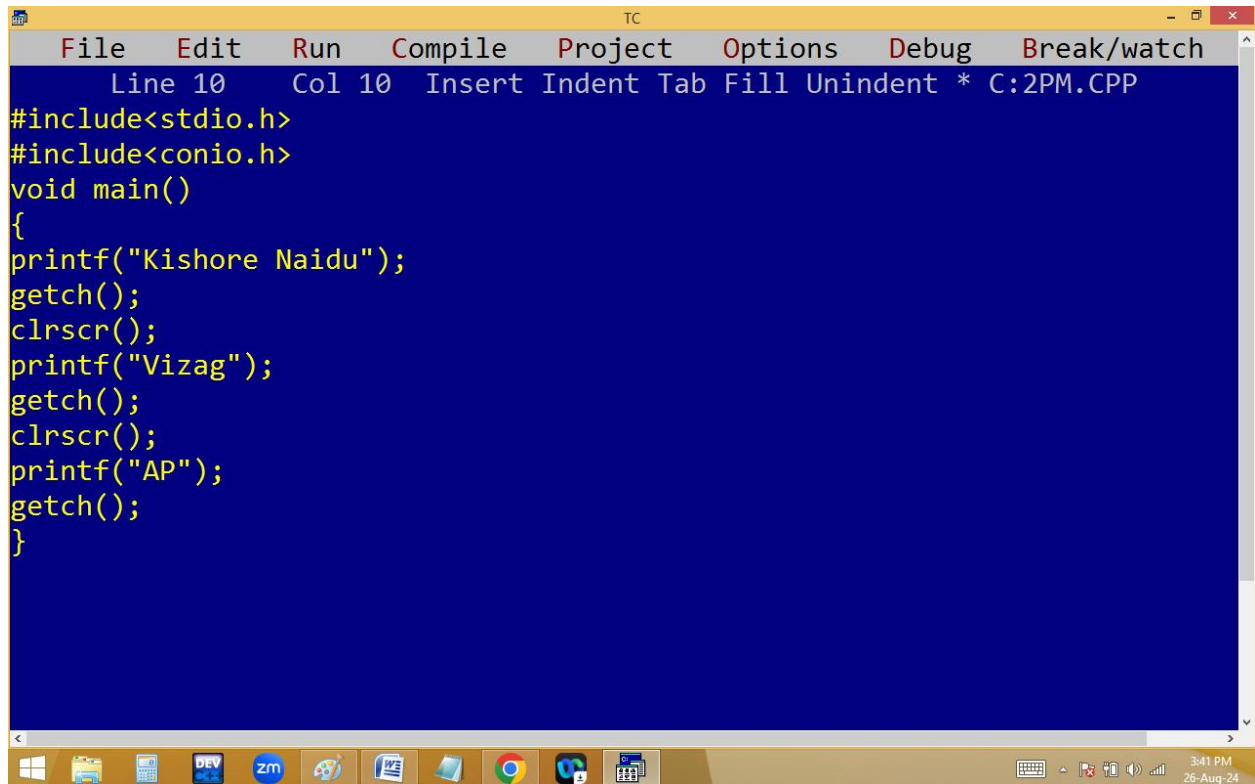
```
File Edit Run Compile Project Options Debug Break/watch
Line 2 Col 10 Insert Indent Tab Fill Unindent * C:2PM.CPP
#include<stdio.h>
#include<conio.h>
void main()
{
printf("Kishore Naidu");
getch();
printf("Vizag");
getch();
printf("AP");
getch();
}
```



```
TC
Kishore Naidu_

```

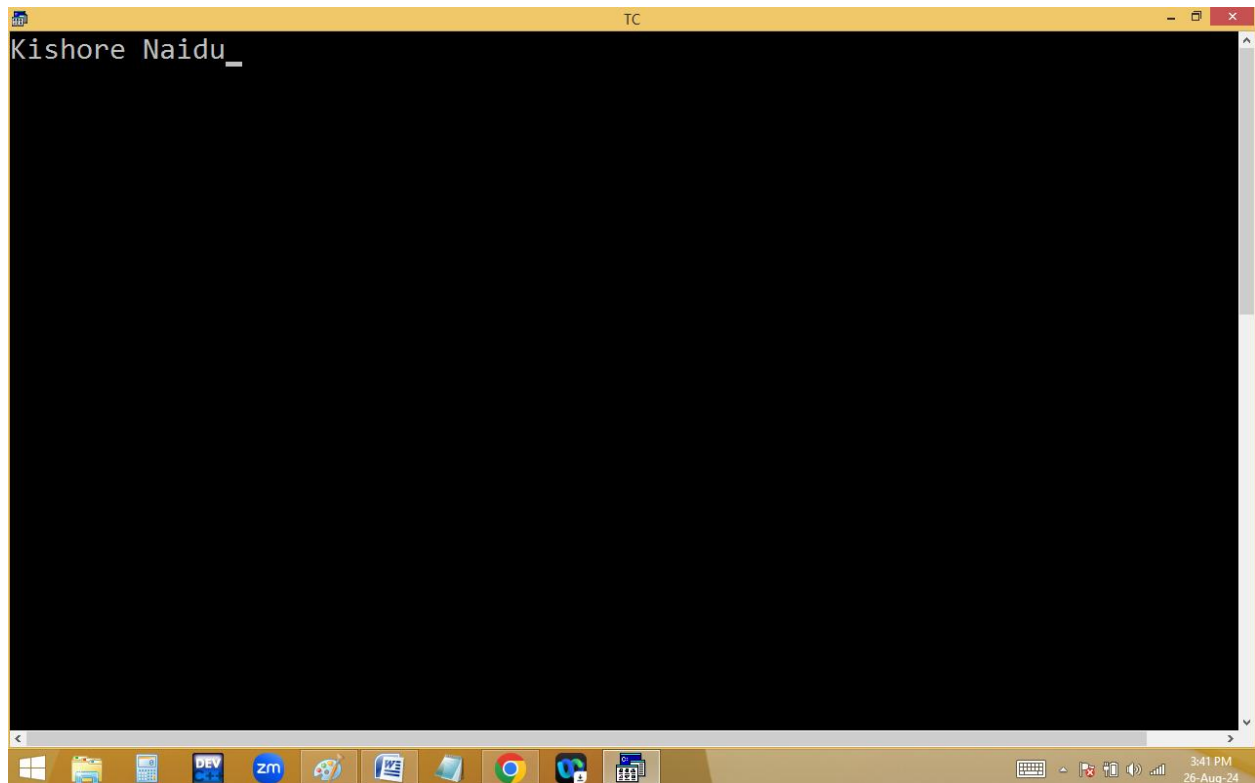




The screenshot shows the Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", "Debug", and "Break/watch". The status bar at the top indicates "Line 10", "Col 10", and "Insert Indent Tab Fill Unindent \* C:2PM.CPP". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
printf("Kishore Naidu");
getch();
clrscr();
printf("Vizag");
getch();
clrscr();
printf("AP");
getch();
}
```

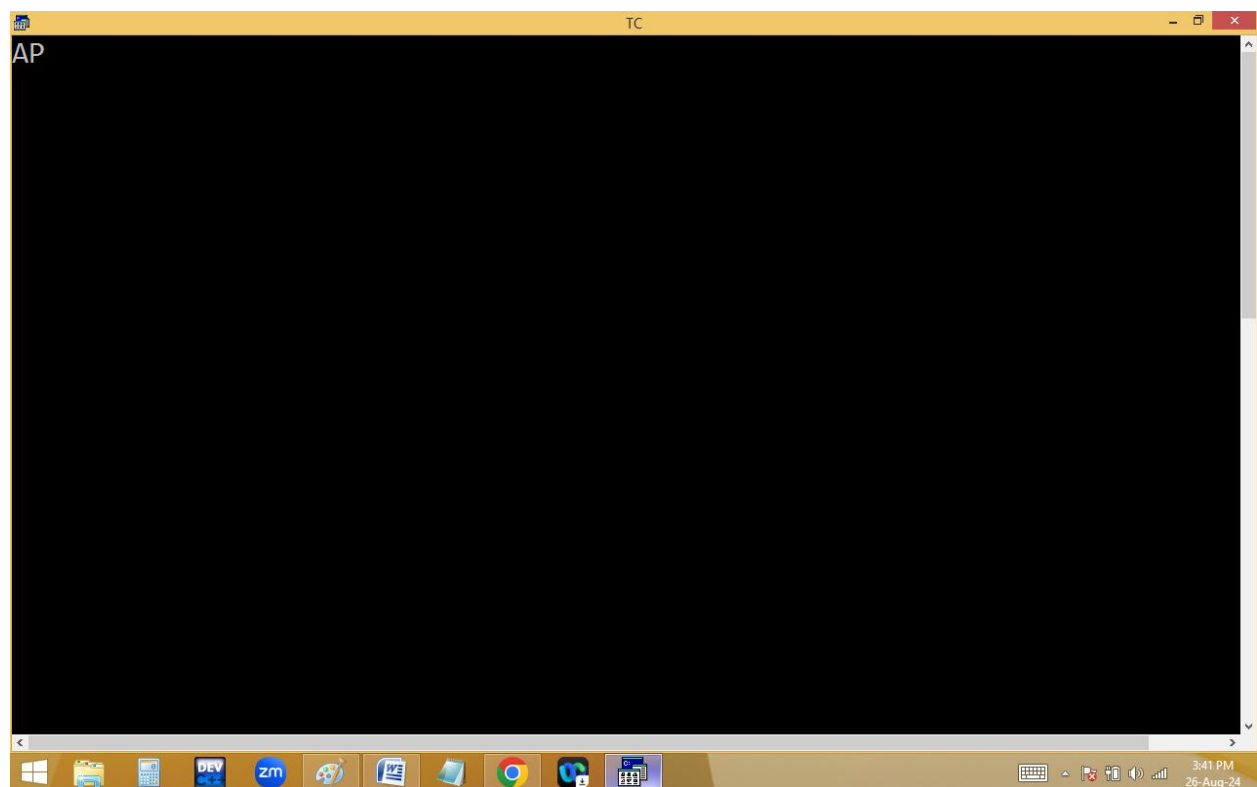
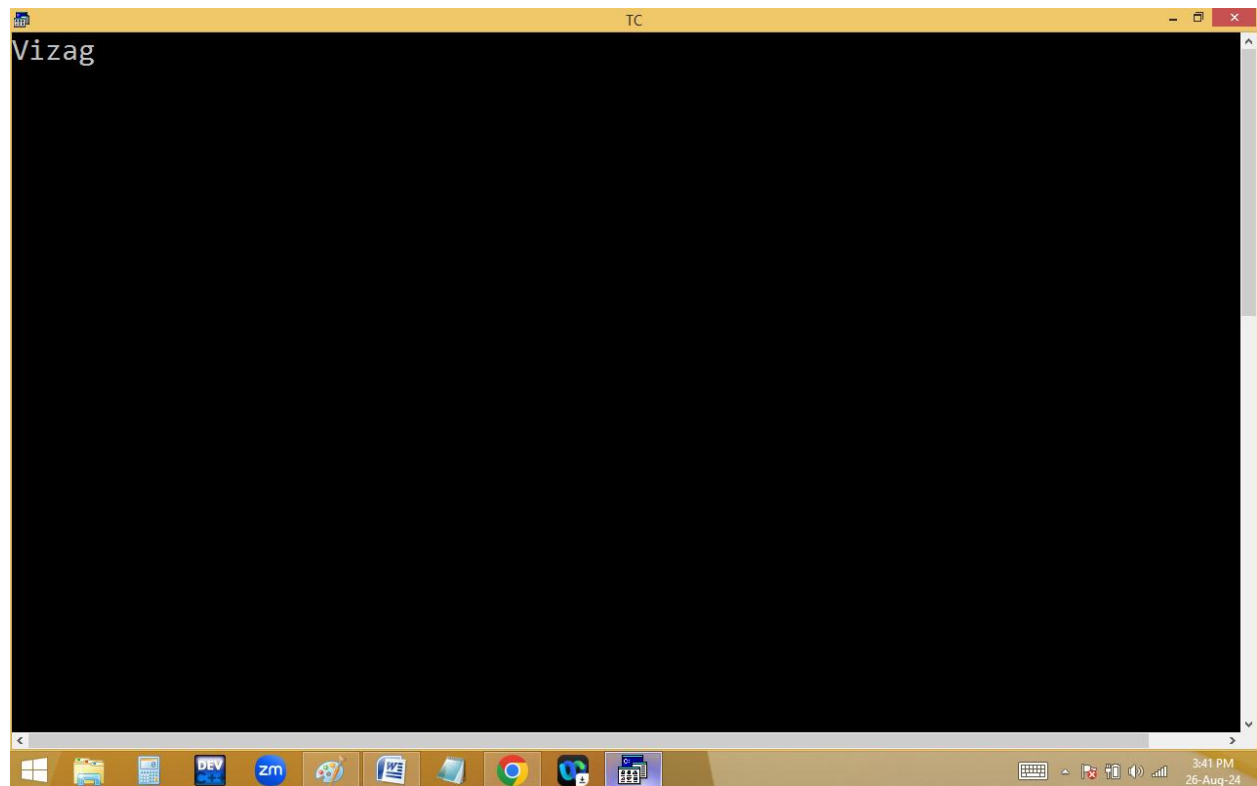
The Windows taskbar at the bottom shows various icons including the Start button, File Explorer, Calculator, DEV, zm, and several other applications. The system clock in the bottom right corner displays "3:41 PM" and "26-Aug-24".



The screenshot shows the Turbo C++ (TC) IDE window after execution. The title bar reads "TC". The main editing area has a black background and displays the output of the program:

```
Kishore Naidu_
```

The Windows taskbar at the bottom is identical to the first screenshot, showing the same set of application icons and system clock information ("3:41 PM", "26-Aug-24").



```
#include<stdio.h>
```

```
#include<conio.h>

#include<stdlib.h>

main()
{
    printf("Kishore");

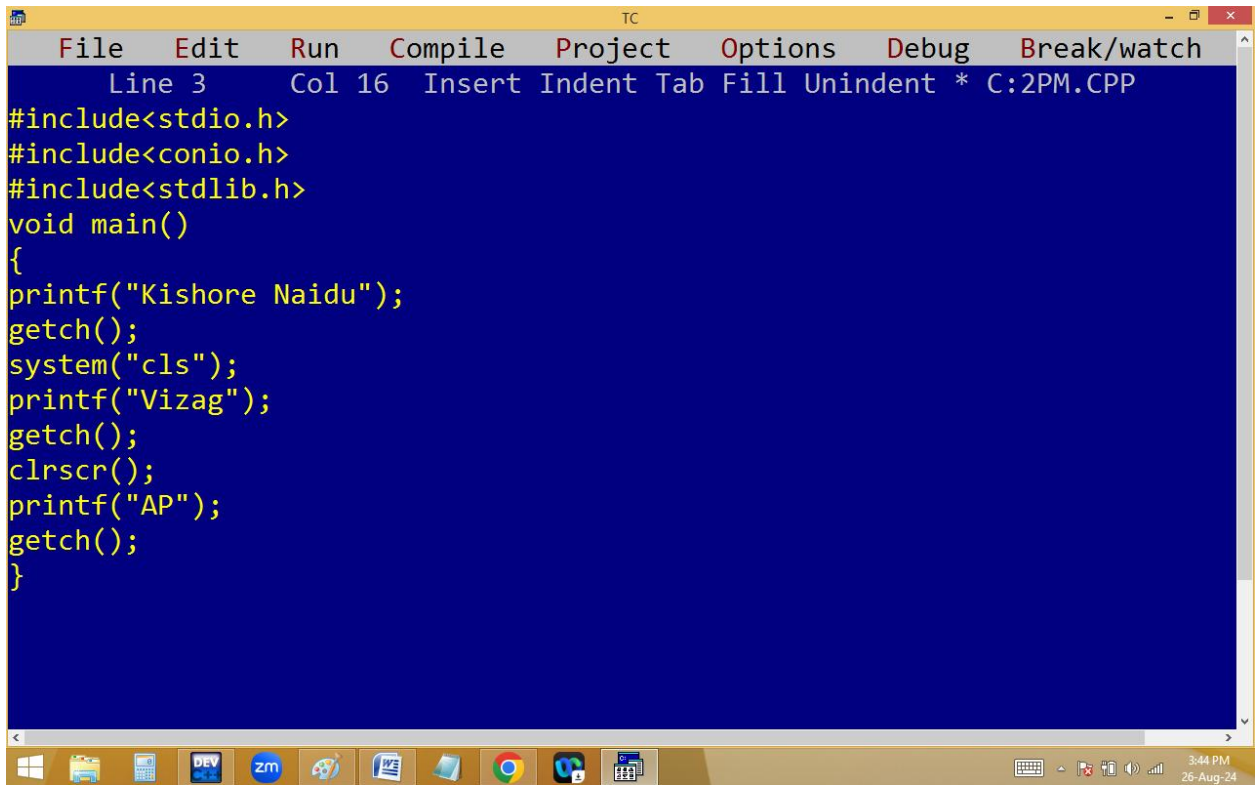
    getch();

    system("cls");

    printf("Naidu");

    getch();

    system("cls");
}
```



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 3 Col 16 Insert Indent Tab Fill Unindent * C:2PM.CPP
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>
void main()
{
printf("Kishore Naidu");
getch();
system("cls");
printf("Vizag");
getch();
clrscr();
printf("AP");
getch();
}
```

## BACK SLASH / ESCAPE SEQUENCE CHARACTERS

They started with back slash [ \ ].

They used to format the outputs.

They participated in program execution but not displayed in output. Hence they are also called **escape sequence characters**.

Each back slash character=**1 byte i.e. one character**.

| BACK SLASH CHARACTER | DESCRIPTION                        |
|----------------------|------------------------------------|
| \a                   | Alert [ beep sound ]               |
| \b                   | Back space                         |
| \n                   | New line character                 |
| \t                   | Tab space                          |
| \r                   | Carriage return[beginning of line] |
| \f                   | Form feed ♀                        |
| \v                   | Vertical tab ♂                     |
| \0                   | Null char                          |
| \\                   | \ [ invalid ]                      |
| \k                   | k [ invalid ]                      |



